MKALAMA DISTRICT COUNCIL

FORM THREE TERMINAL EXAMINATION, MAY 2025

BASIC MATHEMATICS-MARKING SCHEME

1. a) Deposited amount - 360,000/=
   Bank charges per withdrawal =1000/=
   She withdraw 106,000/= two times
   Total amount deducted= 106,000x2+1000x2=214000/= **1.5 marks**
   Remaining amount of money= Tsh 360000-Tsh 214000=Tsh.146, 000/=
   1.5marks

   b) Let $x=2.4\dot{3}$ ----------i

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   Multiply by 10 both side

   $10x=2.4\dot{3} \times 10$

   $10x=24.3\ddot{3}$……ii 1mark

   Subtract equation ii from equation i

   $10x-x=24.3\ddot{3} - 2.4\dot{3}$.....0.5 mark

   $\frac{9x}{x} = \frac{21.9}{9} \times \frac{10}{10} = \frac{219}{90}$ 1mark

   2.433333……..=219/90 $=2\frac{13}{30}$ ..... 0.5 mark

2. a) $4\log 5 - \frac{1}{2}\log 16 + 6\log 2.$

   $= 4\log 5 - \frac{4}{2}\log 2 + 6\log 2.$

   $= 4\log 5 - 2\log 2 + 6\log 2.$ (3 marks)

   $= 4\log 5 + 4\log 2.$

   $= 4(\log(5 \times 2)$

   $= 4(\log(10))$

   $= 4$

   b) $\frac{5^{x+3}}{25^{2x-3}} = 1$

   $\frac{5^{x+3}}{5^{4x-6}} = 1$

   $5^{x+3-(4x-6)} = 1$

   $5^{-3x+9} = 5^0$ (3marks)

   $-3x + 9 = 0$

   $3x = 9$

   $x = 3$

3. (a) Total number of students, μ=100
Students studying physics subject; n (P) =45,
Students studying chemistry n(C) = 40
Students studying both physics and chemistry n (P∩C) = 5
Required: number of students studying none of these subjects
From $n(P \cup C)' = n(\mu) - n(P \cup C)$

$n(P \cup C)' = n(\mu) - (n(P) + n(C) - n(P \cap C)$

$n(P \cup C)' = 100 - (45 + 40 - 5)$ (03marks)

$n(P \cup C)' = 100 - (80$

$n(P \cup C)' = 20$

There are 20 students studying none of these subjects

(b) Three fifth of students in form three class are male and the rest are female. If there are 36 female students, how many male students are in this class?

let x = Total number of students

Male $= \frac{3x}{5}$

Female $= x - \frac{3x}{5} = \frac{2x}{5} = 36$ (03marks)

$x = 90$

Male students $= 90 - 36 = 54$

4. a) Given equation y=3x-p

Points (6, 10) and (q, 22)

From y=3x-p

p=3x-y, substitute (6, 10) 0.5 mark

P=3x6-10=18-10=8

p=8 1mark

Substitute (q, 22) in y=3x-8

22=3q-8 0.5 mark

22+8=3q

30=3q

30/3=q

q=10 1mark

The value of p=8, q=10

b) Given m=3/4, point (-1, 2)

From $m=\frac{\Delta y}{\Delta x} = \frac{y-2}{x--1}$

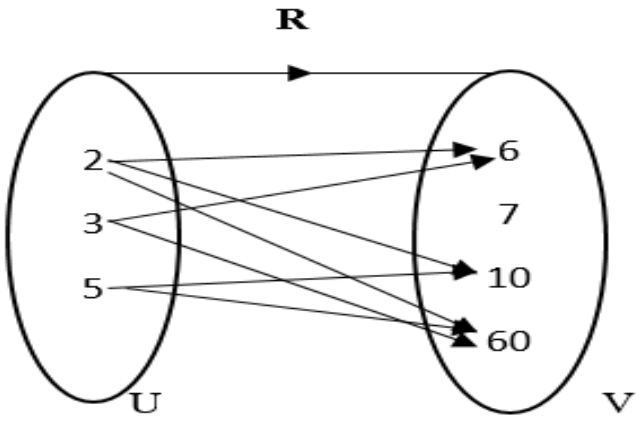
But $m=3/4=\frac{y-2}{x--1}$ 1mark

3x+3=4y-8

4y=3x+11

The equation is $y = \frac{3x}{4} + \frac{11}{4}$ 2mark

5. a)



2marks

b) Given

x+y≤2, x+y=2

Table value

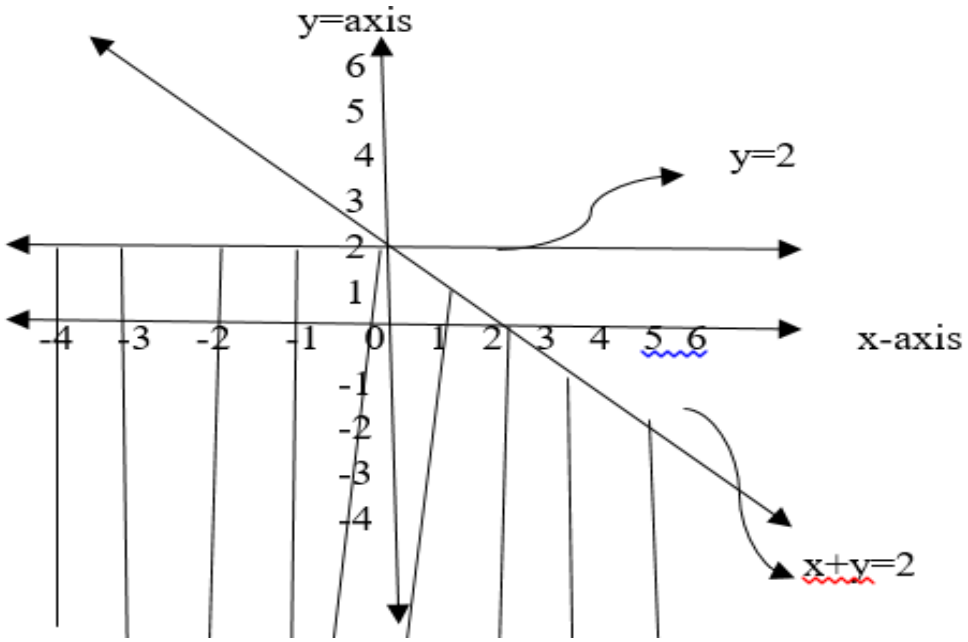
x+y=2=y=2-x

x	-2	-1	0	1	2
y	4	3	2	1	0

1/2mark

Given y≤ 2 = $y = 2$ 1/2mark

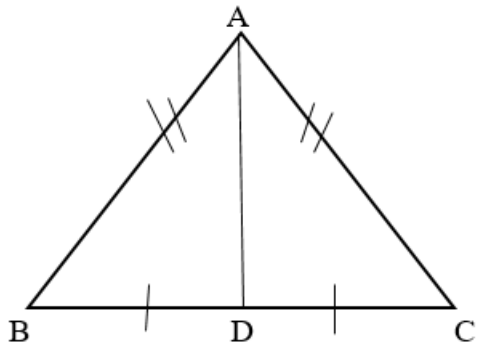
Graph..... (2marks)



Domain{x: x∈ *R*} 1/2mark

Range {y: y≤ 2} 1/2mark

6. a) Consider the figure below



Proof

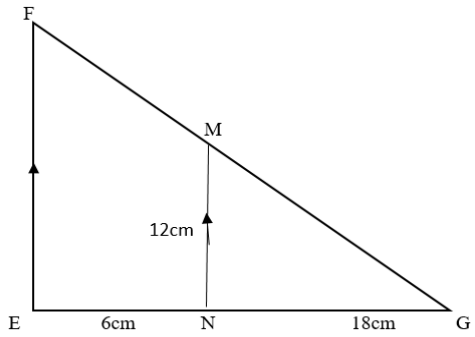
AD is common side (s) 0.5mark

AB=AC Given sides (s) 0.5mark

BD=DC Given sides (s) 0.5mark

$\Delta ABD \cong \Delta ACD \; by \; sss \; rule$ 0.5marks

b) Consider the figure below



FE//MN Given side (s) 0.5mark

EG//NG Given side (s) 0.5mark

FG//MG Given side (s) 0.5mark

The pairs of similar triangles are $\Delta FEG \; and \; \Delta MNG$

From the ratio of corresponding sides

$\frac{FE}{MN} = \frac{EG}{NG} = \frac{FG}{MG}$ 0.5mark

$\frac{FE}{12cm} = \frac{6cm+18cm}{18cm}$ 1mark

$$\frac{FE}{12cm} = \frac{24cm}{18cm}$$

FE=$\frac{24cmx12cm}{18cm}$

FE=$\frac{288cm2}{18cm} = 16cm$

The length $\overline{FE}$ is 16cm 01mark

7. a) $x \propto 2y + 7$ 0.5mark

Given x1=5 y1=4

X2=6 y2=?

$\frac{x}{2y+7} = k\left(\frac{2y+7}{2y+7}\right)$

k=$\frac{x}{2y+7}$ 0.5mark

$\frac{x1}{2y1+7} = \frac{x2}{2y2+7}$

$\frac{5}{2(4)+7} = \frac{6}{2y2+7}$

$\frac{5}{8+7} = \frac{6}{2y2+7}$

$\frac{5}{15} = \frac{6}{2y2+7}$

5(2y2+7)=6(15)

10y2+35=90

10y2=90-35

10y2=55

Divide by 10 both side

10y2/10=55/10=5.5

y2=5.5

The value of y=5.5 2marks

b) $W\ \alpha\ \frac{1}{h}\ \ and\ W\alpha\frac{1}{d}$

Where by

w=workers

h= hours

d= days

Joint variation

$w = \frac{k}{hd}$

k=whd

w1h1d1=w2h2d2 0.5marks

d1=52 , h1=8 ,w1=9

w2=13 and h2=6 d2=?

d2= $\frac{w1h1d1}{w2h2}$

d2=$\frac{9\times8\times52}{13\times6} = 48days$

It will complete 48days 2marks

8. (a) from nth term=2^{n-1}

5th term=2^4

10th term=2^9

Sum=16+512=528

(b) 1985, 1986, 1987... 2000

$from\ An = A1 + (n-1)d$ (1.5 marks)

$$2000 = 1985 + (n-1) \times 1$$

$$n = 16$$

Coconut produced 10+13+16+19+.....

From $S_n = \frac{n}{2}(2A1 + (n-1)d$

$S16 = \frac{16}{2}(2 \times 10 + (16-1) \times 3)$ (1.5 marks)

$$\underline{S16 = 520}$$

<u>Coconuts produced from 1985 to 2000 is 520 coconuts</u>

9. a) Given function f(x)=x^2+6x-7

a=1 , b=6 , c=-7

i) AxB of symmetry X=-b/2a=-6/2=-3 1mark

ii) $y = \frac{4ac-b^2}{4a} = \frac{-28-36}{4} = -16$ (2marks)

Maximum value is -16

(b) $x^2 + 6x - 7 = 0$

$x^2 + 6x = 7$

$(x+3)^2 - 9 = 7$ (1.5 marks)

$(x+3)^2 = 16$

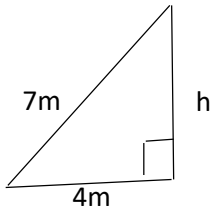
By applying square root both sides

$\sqrt{(x+3)^2} = \mp\sqrt{16}$

$x + 3 = \mp 4$ (1.5 marks)

$\underline{Either\ x = -7\ or\ x = 1}$

10. (a)



$from\ a^2 + b^2 = c^2$ (03marks)

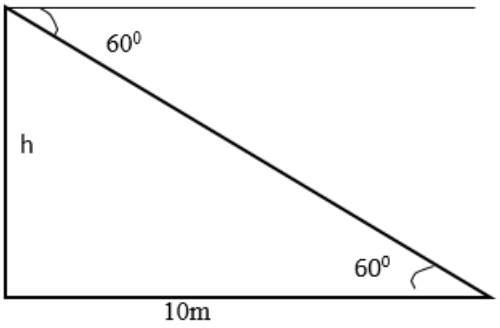
$4^2 + h^2 = 7^2$

$h = \sqrt{33}m$

$h \approx 5.74m$

The point where the ladder meets the wall is approximately 5.74 m high above the floor

b)



1mark

From tan 60^0=opposite/adjacent 0.5mark

Tan 60^0=h/10m 0.5mark

h=tan60^0x10m

h=10mx1.732 =17.32m 0.5mark

The lower is 17.32m 0.5mark

11. a) given A=34.5

Intervals	f	x	d=x-A	fd
0-9	3	4.5	-30	-90
10-19	4	14.5	-20	-80
20-29	6	24.5	-10	-60
30-39	8	34.5	0	0
40-49	3	44.5	10	30
50-59	5	54.5	20	100
60-69	3	64.5	30	90
70-79	2	74.5	40	80
80-89	1	84.5	50	50

4mark

$\varepsilon f = 35$ $and\ \varepsilon fd = 120$

b) Median =L+ $(\frac{\frac{N}{2}-Nb}{Nw})i$ 1mark

L=29.5
N=35
Nb=13
i=10
Nw=8

Median=29.5+$(\frac{\frac{35}{2}-13}{8})10$

29.5+5.625=35.125=35.13

The median =35.13 2marks

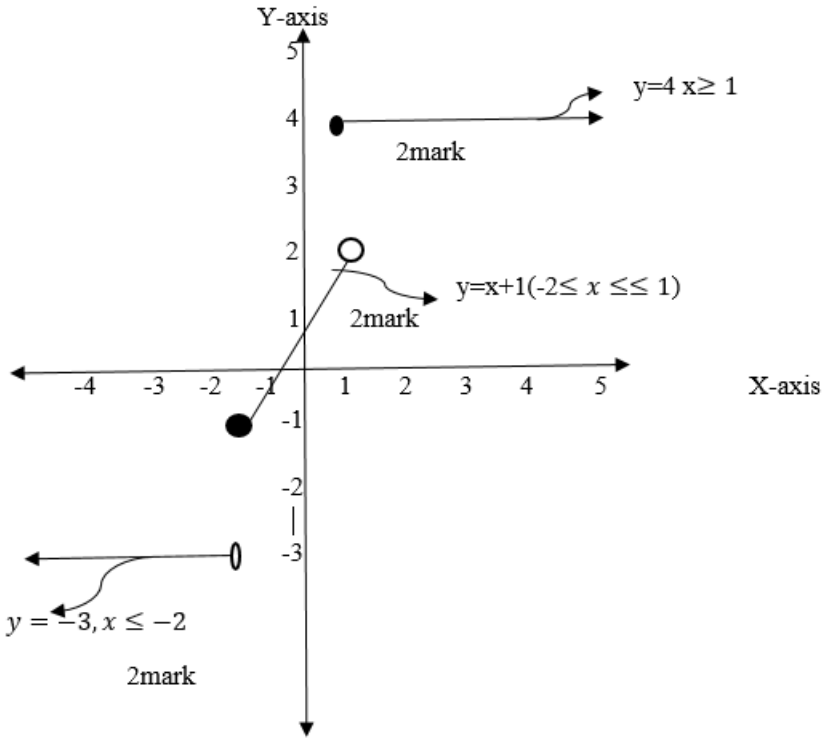
c) Mean=A+$\frac{\varepsilon fd}{\varepsilon f}$ 1mark

34.5+120/35

34.5+3.4285=37.93

The mean =37.93 2marks

12. i) Drawn graph



y=x-1

Table value

x	-2	-1	0	2
y	-1	0	1	2

1mark

ii)g(-5)=-3 1mark

g(10)=4 1mark

i) Domain={x:x∈ R} 1/2mark

Range={y:y=-3,-1,0,1,2,4} 1/2mark

13. a)Require to factorize p^2-q^2

By using different of two squares

$p^2-q^2= (p+q)(p-q)$

Then $p^2-q^2= (p+q)(p-q)$ 3mark

b) From 12a above $p^2-q^2= (p+q)(p-q)$

Then $1024-64=32^2-8^2$ 1mark

Since $p^2-q^2=(p+q)(p-q)$

$32^2-8^2=(32+8)(32-8)$ 2mark

$=(40)(24)=960$

Then 1024-64=960 1mark

c) Given $6x^2-11xy-y^2$

Find factors of 60 from

$6x^2-11xy-10y^2$ 1mark

Factors are -15 and 4 then

$6x^2-15xy+4xy-10y^2$ 1mark

$(6x^2-15xy)+(4xy-10y^2)$

$3x(2x-5y)+2y(2x-5y)$ 1mark

$(3x+2y)(2x-5)$

$6x^2-11xy-10y^2=(3x+2y)(2x-5y)$ 1mark

14. a) $y>x$

i) The ordered pairs belong to relation $R=\{X,Y:Y>X\}$ are

(1,2) ,(-3,4) ,(-5,0), (-8,-3) 1mark

ii) Domain= {1,-3,-5,-8} 2marks

Range ={2,4,0,-3} 2marks

b) Rate of growth of seedling A

5cm/10days=0.5cm/day 2marks

Rate of growth of seedling B

8cm/12days=0.67cm/day 2marks

The growth rate of seedling B was greater than seedling A hence seedling B was growing more quickly than the seedling A 1mark